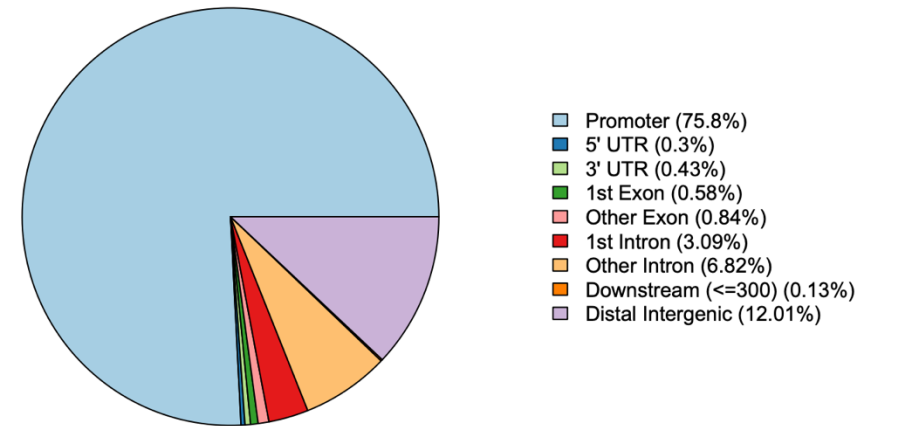

Summary of TCF4 project

The background features a complex network of nodes and lines. Nodes are represented by small circles, some of which are highlighted with a white outline. Lines connect these nodes, forming a web-like structure. The lines are thin and have a color gradient, ranging from light blue to yellow. The overall composition is abstract and technical, suggesting a network or data structure.

Peak calling, MEME-ChIP Analysis, and

- After IDR process → 3,947 peaks using hg19 → 3,945 peaks using hg38
- Top enriched motifs: SP1/SP2/SP3, NRF1, NFY (centrally enriched)
- 75.8% of your peaks fall within ±1 kb of Transcription Start site (TSS)
- 77.02% of peaks contain an E box motif hit



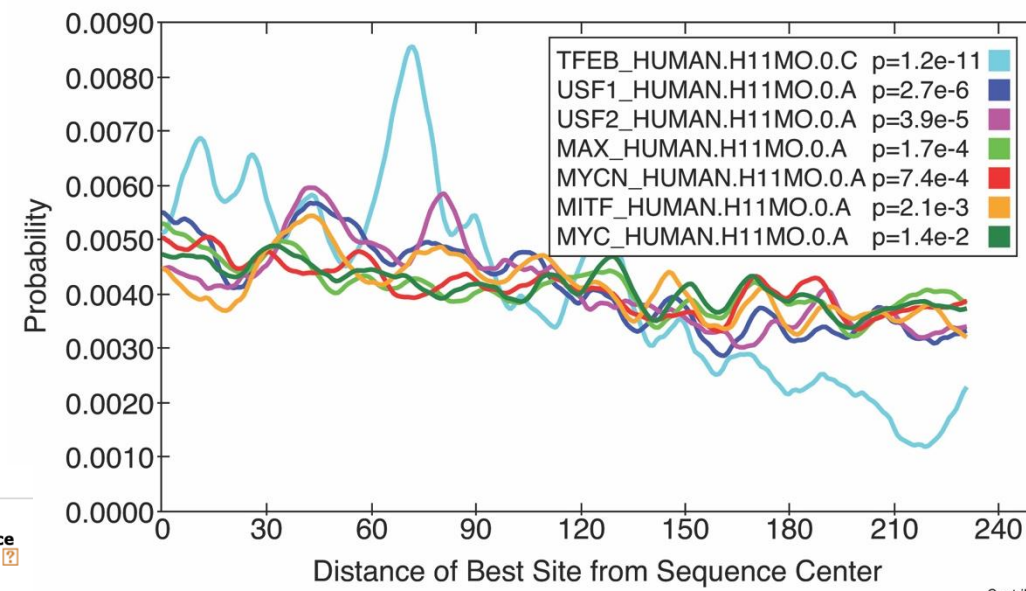
85.2% of non-promoter peaks contain E-box
74.4% of promoter peaks contain E-box

↓

E-boxes were significantly **less frequent in promoter regions** compared to non-promoter regions

CentriMo enriched motif summary

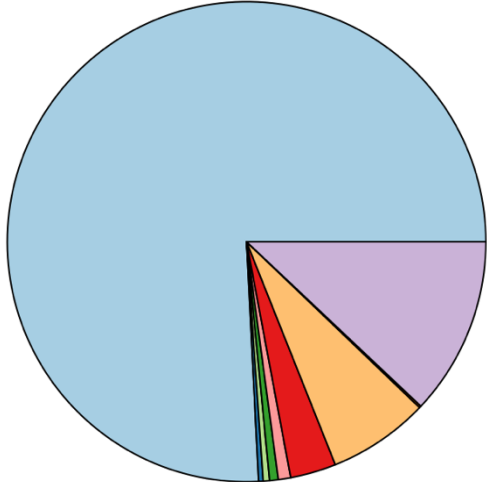
- E-box–family motifs are significantly enriched at peak centers.



Enriched motifs (E-value ≤ 10 using the binomial test)

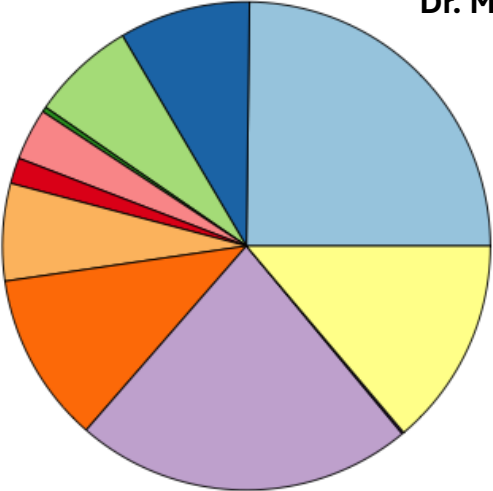
☒	ID ?	Alt ID ?	Consensus ?	Concentration ?	E-value ?	p-value ?	Region Width ?	Region Matches ?	Sequence Matches ?
■	TFEB_HUMAN.H11MO.0.C		RGTCACGTG	0.0612	5.2e-9	1.2e-11	268	279	381
■	USF1_HUMAN.H11MO.0.A		RGTCACGTGRBS	0.0575	1.1e-3	2.7e-6	219	498	922
■	USF2_HUMAN.H11MO.0.A		GGGTCACGTGRBSVSSSSB	0.0430	1.6e-2	3.9e-5	230	582	1046
■	MAX_HUMAN.H11MO.0.A		VRGCACGTGG	0.0533	7.3e-2	1.7e-4	91	707	3223
■	MYCN_HUMAN.H11MO.0.A		RGCCACGTGSDS	0.0501	3.1e-1	7.4e-4	125	823	2803
■	MITF_HUMAN.H11MO.0.A		TCACGTGACC	0.0425	8.7e-1	2.1e-3	265	968	1632
■	MYC_HUMAN.H11MO.0.A		RCCACGTGCTB	0.0491	6.1	1.4e-2	274	1521	2547

NPC TCF4 peaks



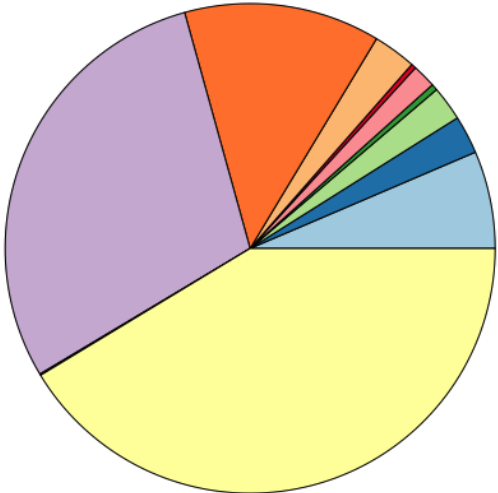
- Promoter (75.8%)
- 5' UTR (0.3%)
- 3' UTR (0.43%)
- 1st Exon (0.58%)
- Other Exon (0.84%)
- 1st Intron (3.09%)
- Other Intron (6.82%)
- Downstream (<=300) (0.13%)
- Distal Intergenic (12.01%)

Dr. McClay Consensus TCF4 peaks



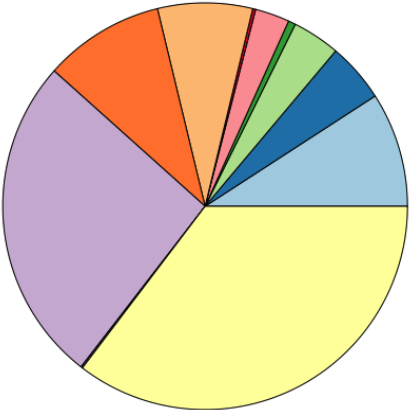
- Promoter (<=1kb) (24.8%)
- Promoter (1-2kb) (8.69%)
- Promoter (2-3kb) (6.9%)
- 5' UTR (0.34%)
- 3' UTR (3.42%)
- 1st Exon (1.72%)
- Other Exon (6.45%)
- 1st Intron (11.29%)
- Other Intron (22.43%)
- Downstream (<=300) (0.12%)
- Distal Intergenic (13.84%)

Forrest TCF4 Peaks



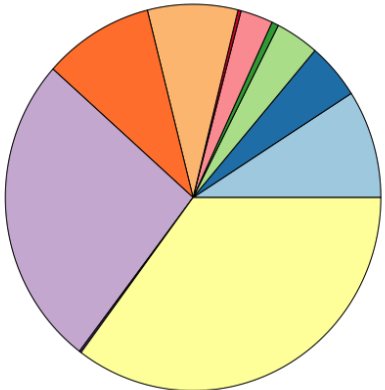
- Promoter (<=1kb) (6.43%)
- Promoter (1-2kb) (2.52%)
- Promoter (2-3kb) (2.29%)
- 5' UTR (0.41%)
- 3' UTR (1.56%)
- 1st Exon (0.32%)
- Other Exon (2.84%)
- 1st Intron (12.94%)
- Other Intron (29.27%)
- Downstream (<=300) (0.0)
- Distal Intergenic (41.33%)

K12 TCF4 Peaks



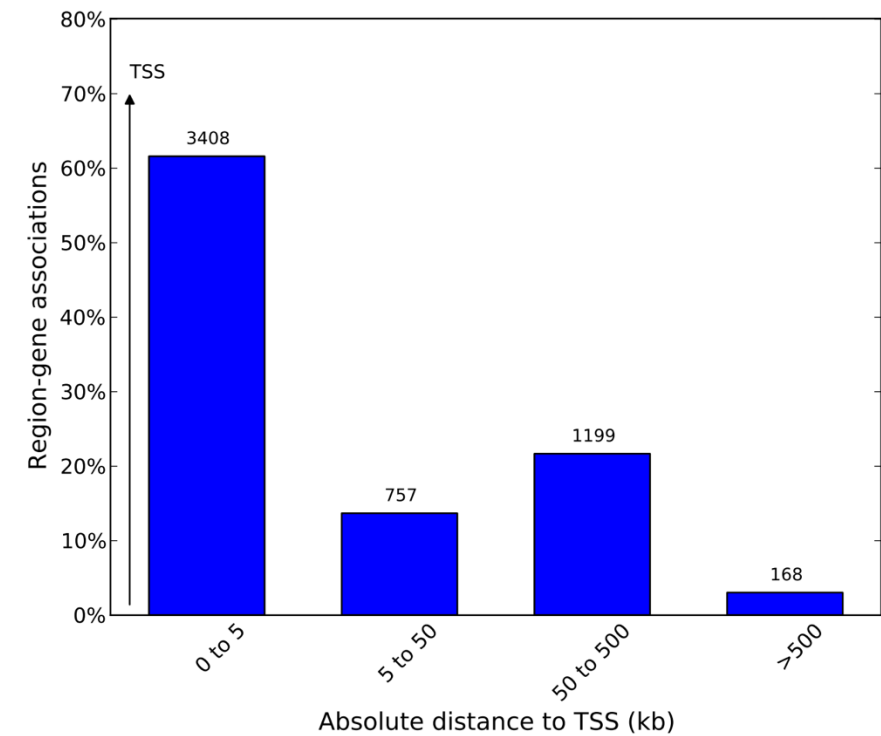
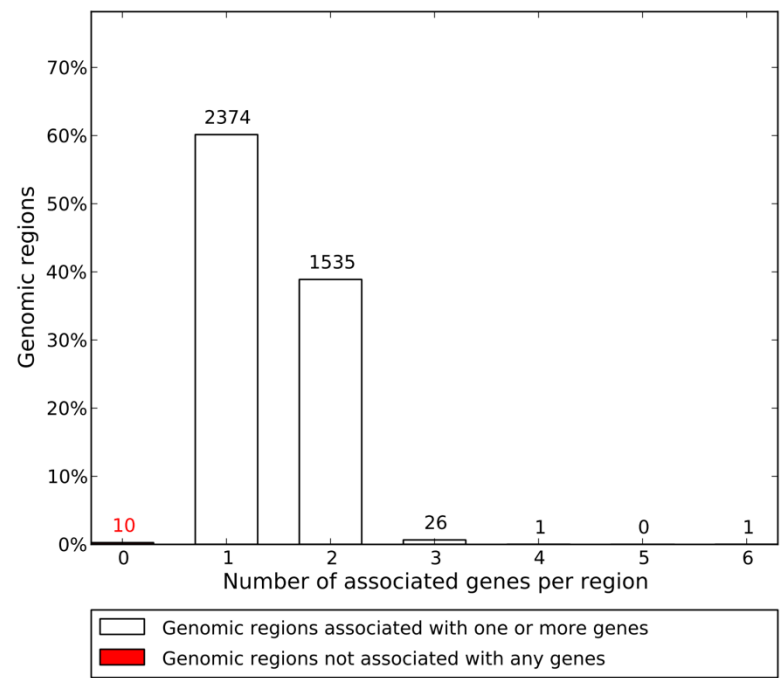
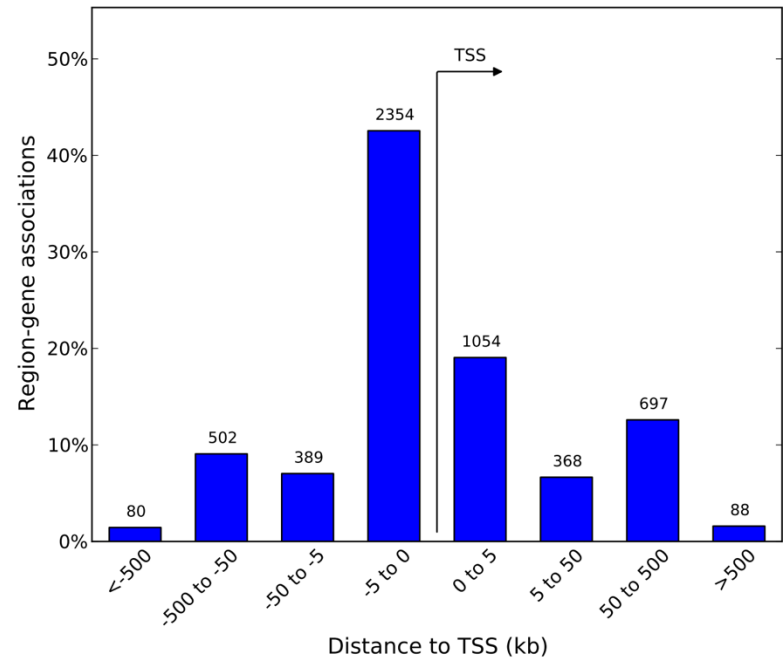
- Promoter (<=1kb) (9.12%)
- Promoter (1-2kb) (4.71%)
- Promoter (2-3kb) (3.81%)
- 5' UTR (0.58%)
- 3' UTR (2.75%)
- 1st Exon (0.24%)
- Other Exon (7.57%)
- 1st Intron (9.62%)
- Other Intron (26.09%)
- Downstream (<=300) (0.14)
- Distal Intergenic (35.37%)

N16 TCF4 Peaks

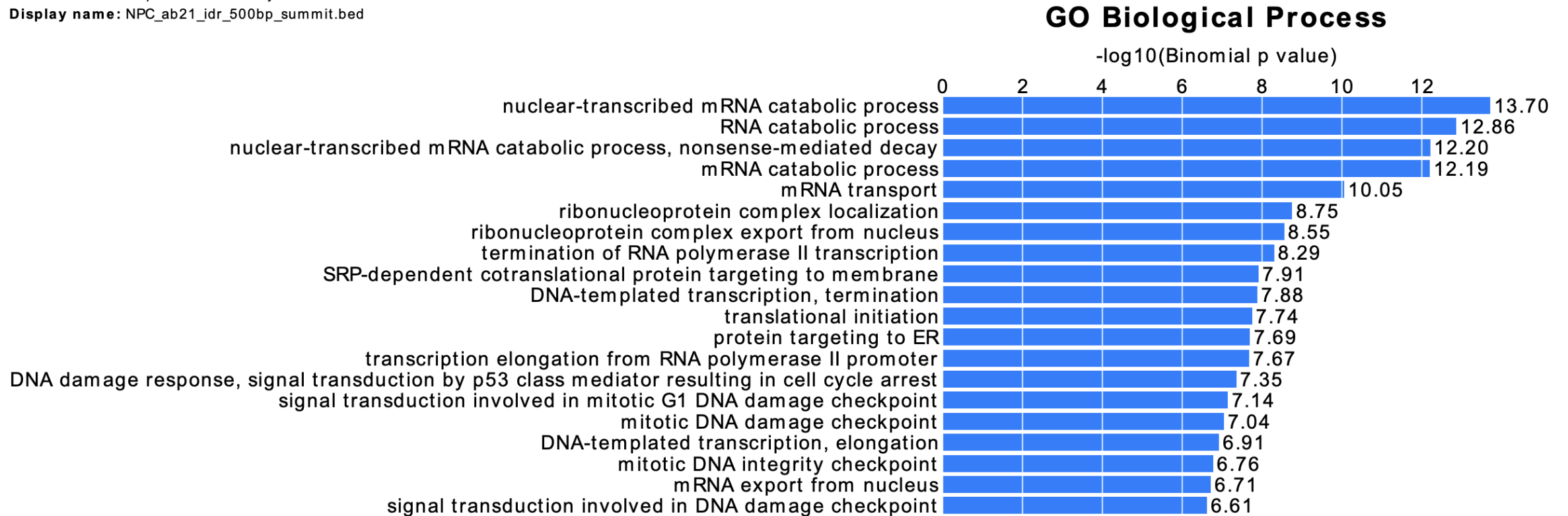


- Promoter (<=1kb) (9.06%)
- Promoter (1-2kb) (4.7%)
- Promoter (2-3kb) (3.74%)
- 5' UTR (0.59%)
- 3' UTR (2.79%)
- 1st Exon (0.27%)
- Other Exon (7.77%)
- 1st Intron (9.5%)
- Other Intron (26.24%)
- Downstream (<=300) (0.17%)
- Distal Intergenic (35.18%)

GREAT Results (All 3947 peaks are included)



Display name: NPC_ab21_jdr_500bp_summit.bed

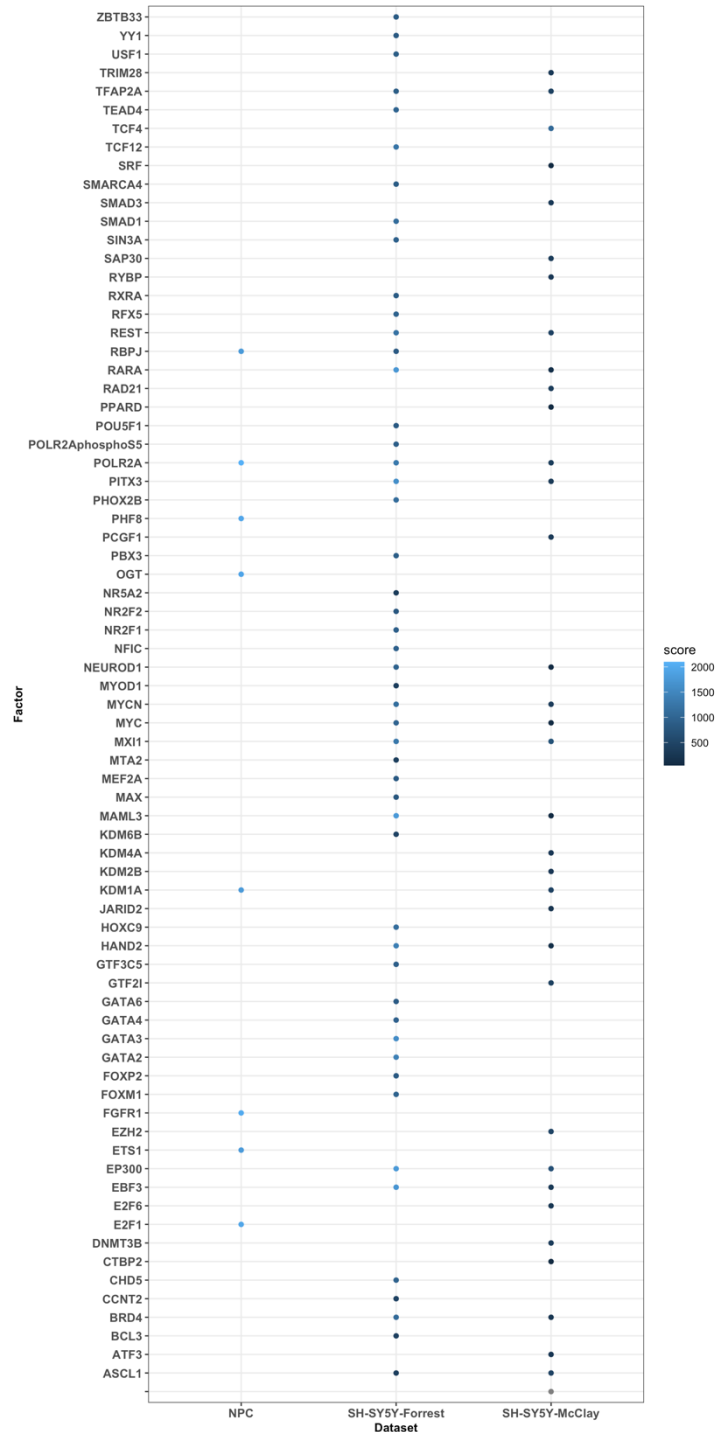
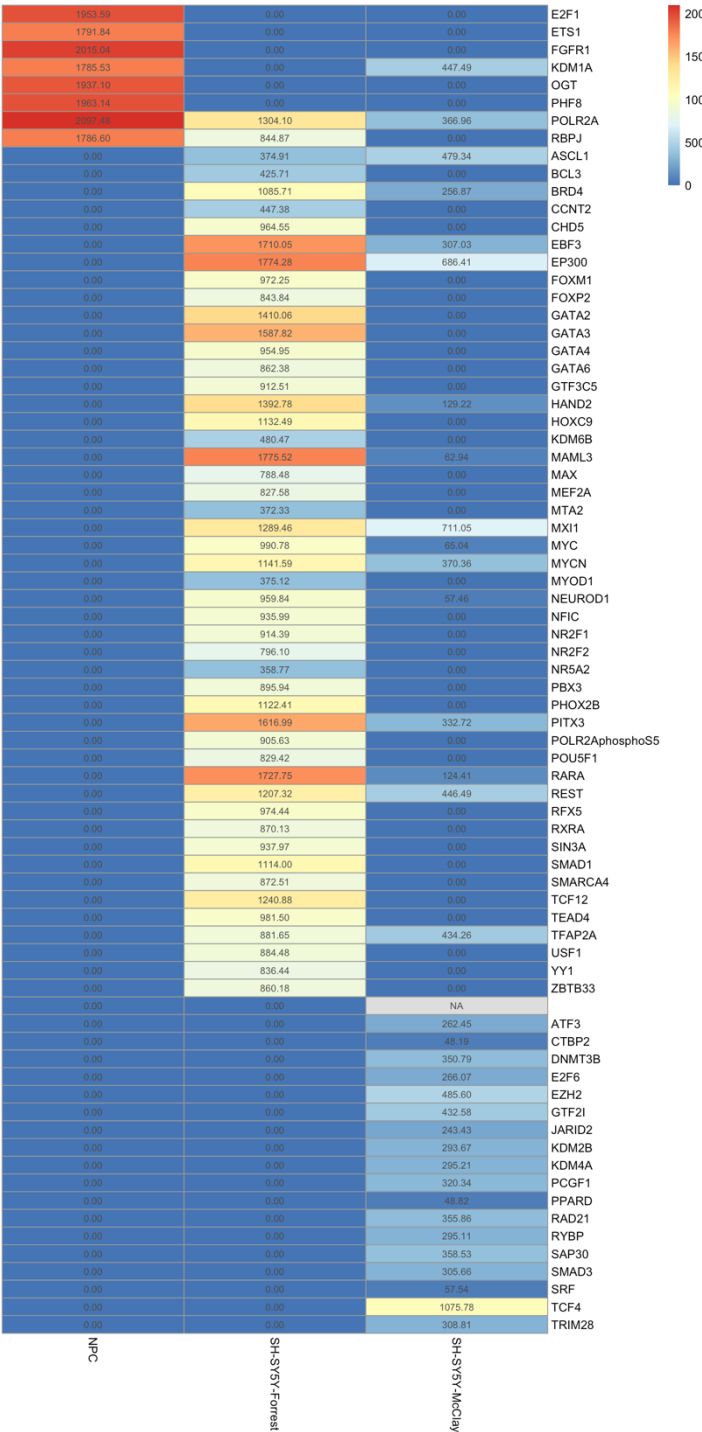


Ontology	# Term Name	Binom Rank	Binom Raw P-Value	Binom FDR Q-Val	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	Hyper FDR Q-Val	Hyper Fold Enrichment	Hyper Observed Gene Hits	Hyper Total Genes	Hyper Gene Set Coverage
GO Biological Process	nuclear-transcribed mRNA catabolic process	12	1.97983e-14	2.16873e-11	2.3335	103	2.61%	248	3.91819e-5	1.6343	80	197	1.74%
	RNA catabolic process	13	1.37632e-13	1.39167e-10	2.1006	121	3.07%	262	5.90700e-5	1.5595	93	240	2.02%
	nuclear-transcribed mRNA catabolic process, nonsense-mediated decay	15	6.29055e-13	5.51262e-10	2.8374	64	1.62%	274	1.01656e-4	1.7924	53	119	1.15%
	mRNA catabolic process	16	6.45238e-13	5.30103e-10	2.1712	106	2.69%	284	1.24184e-4	1.5756	83	212	1.80%
	mRNA transport	31	8.87426e-11	3.76297e-8	2.1845	85	2.15%	180	6.66950e-7	1.8744	68	146	1.48%
	ribonucleoprotein complex localization	42	1.78747e-9	5.59436e-7	2.3070	65	1.65%	188	1.69763e-6	1.9164	60	126	1.30%
	ribonucleoprotein complex export from nucleus	44	2.84853e-9	8.50998e-7	2.3132	63	1.60%	208	5.75626e-6	1.8824	58	124	1.26%
	termination of RNA polymerase II transcription	49	5.09632e-9	1.36717e-6	3.0288	38	0.96%	342	3.82656e-4	1.9831	34	69	0.74%
	SRP-dependent cotranslational protein targeting to membrane	55	1.23272e-8	2.94621e-6	2.6085	46	1.17%	720	3.86785e-2	1.5579	36	93	0.78%
	DNA-templated transcription, termination	57	1.33335e-8	3.07489e-6	2.3878	54	1.37%	493	6.69495e-3	1.6246	44	109	0.95%
	translational initiation	59	1.80937e-8	4.03121e-6	2.1381	67	1.70%	410	2.06781e-3	1.5930	57	144	1.24%
	protein targeting to ER	61	2.03451e-8	4.38420e-6	2.4774	49	1.24%	703	3.46113e-2	1.5388	39	102	0.85%
	transcription elongation from RNA polymerase II promoter	63	2.13856e-8	4.46213e-6	2.5006	48	1.22%	510	9.46733e-3	1.6522	39	95	0.85%
	DNA damage response, signal transduction by p53 class mediator resulting in cell cycle arrest	70	4.49402e-8	8.43912e-6	2.5545	44	1.11%	177	4.35751e-7	2.4017	37	62	0.80%
	signal transduction involved in mitotic G1 DNA damage checkpoint	73	7.26168e-8	1.30760e-5	2.5100	44	1.11%	181	7.85571e-7	2.3636	37	63	0.80%
	mitotic DNA damage checkpoint	75	9.21961e-8	1.61589e-5	2.1260	61	1.55%	196	3.30282e-6	2.0123	50	100	1.08%
	DNA-templated transcription, elongation	78	1.22582e-7	2.06582e-5	2.2858	51	1.29%	711	3.62481e-2	1.5092	42	112	0.91%
	mitotic DNA integrity checkpoint	83	1.71793e-7	2.72075e-5	2.0585	63	1.60%	207	5.63657e-6	1.9558	52	107	1.13%
	mRNA export from nucleus	87	1.94441e-7	2.93785e-5	2.2300	52	1.32%	293	1.60230e-4	1.8224	48	106	1.04%
	signal transduction involved in DNA damage checkpoint	88	2.45301e-7	3.66419e-5	2.3715	45	1.14%	179	6.01631e-7	2.3528	38	65	0.82%
GO Cellular Component	catalytic step 2 spliceosome	25	5.60735e-10	3.87132e-8	2.5485	57	1.44%	40	1.84033e-8	2.2029	52	95	1.13%
	methyltransferase complex	30	3.71083e-8	2.13497e-6	2.3124	54	1.37%	66	2.06859e-4	1.8575	42	91	0.91%
	nuclear ubiquitin ligase complex	33	4.70027e-7	2.45839e-5	3.2846	25	0.63%	95	9.41858e-3	2.0123	20	40	0.43%
	spliceosomal snRNP complex	35	1.78575e-6	8.80629e-5	2.6616	31	0.79%	68	2.78107e-4	2.0452	31	61	0.67%
	histone methyltransferase complex	36	1.99980e-6	9.58793e-5	2.2106	44	1.11%	74	6.84841e-4	1.9248	33	69	0.72%
	small nuclear ribonucleoprotein complex	42	9.19856e-6	3.78017e-4	2.3696	33	0.84%	70	4.95130e-4	1.9531	33	68	0.72%
	cyclin-dependent protein kinase holoenzyme complex	51	7.20530e-5	2.43850e-3	2.3345	27	0.68%	107	1.72103e-2	1.9607	19	39	0.41%
	transcription elongation factor complex	55	9.59052e-5	3.00968e-3	2.0940	33	0.84%	78	1.40812e-3	1.9770	28	57	0.61%
	NELF complex	57	1.06990e-4	3.23975e-3	11.0817	5	0.13%	134	4.90519e-2	4.0245	4	4	0.09%
	precatalytic spliceosome	68	2.88031e-4	7.31090e-3	3.0655	14	0.35%	132	4.62355e-2	2.0123	14	28	0.30%
	tRNA-intron endonuclease complex	69	3.25018e-4	8.13017e-3	12.6340	4	0.10%	134	4.90519e-2	4.0245	4	4	0.09%
	PTW/PP1 phosphatase complex	85	1.01976e-3	2.07071e-2	4.5859	7	0.18%	110	2.03002e-2	3.4496	6	7	0.13%
	MIS12/MIND type complex	89	1.15607e-3	2.24200e-2	8.9678	4	0.10%	134	4.90519e-2	4.0245	4	4	0.09%
	preribosome, large subunit precursor	94	1.27204e-3	2.33568e-2	2.6228	14	0.35%	112	2.19431e-2	2.1670	14	26	0.30%
	spliceosomal tri-snRNP complex	97	1.51747e-3	2.70016e-2	2.5726	14	0.35%	132	4.62355e-2	2.0123	14	28	0.30%
	transcription factor AP-1 complex	104	2.10748e-3	3.49761e-2	4.6668	6	0.15%	102	1.60016e-2	4.0245	5	5	0.11%
GO Molecular Function	cyclin-dependent protein serine/threonine kinase regulator activity	7	6.74507e-7	4.06535e-4	3.1312	26	0.66%	68	7.21951e-4	2.5872	18	28	0.39%
	unfolded protein binding	8	1.35518e-6	7.14690e-4	2.2696	43	1.09%	134	2.79955e-2	1.5715	41	105	0.89%
	telomeric DNA binding	28	2.07848e-4	3.13182e-2	2.4099	22	0.56%	142	3.99721e-2	2.0123	17	34	0.37%
	cyclin-dependent protein serine/threonine kinase activity	29	2.33128e-4	3.39162e-2	2.3895	22	0.56%	80	2.38292e-3	2.4147	18	30	0.39%
	cyclin-dependent protein kinase activity	31	2.66600e-4	3.62834e-2	2.3657	22	0.56%	86	4.05935e-3	2.3368	18	31	0.39%
	cullin family protein binding	35	4.16869e-4	5.02506e-2	3.2854	12	0.30%	141	3.50406e-2	2.4594	11	18	0.24%
Mouse Phenotype	abnormal DNA repair	5	1.59695e-8	3.05146e-5	2.2053	63	1.60%	211	9.75203e-5	1.7813	54	122	1.17%

The test set of 3,947 genomic regions picked 4,609 genes (25%) of all 18,549 genes.	
GO Biological Process has 13,145 terms covering 16,621 (90%) of all 18,549 genes.	13,145 ontology terms were tested (100%) using an annotation count range of [1, 1000].
GO Cellular Component has 1,726 terms covering 17,726 (96%) of all 18,549 genes.	1,726 ontology terms were tested (100%) using an annotation count range of [1, 1000].
GO Molecular Function has 4,219 terms covering 16,545 (89%) of all 18,549 genes.	4,219 ontology terms were tested (100%) using an annotation count range of [1, 1000].
Mouse Phenotype has 9,554 terms covering 9,592 (52%) of all 18,549 genes.	9,554 ontology terms were tested (100%) using an annotation count range of [1, 1000].

Cistrome results

NPC-only factor	Possible meaning
E2F1	Cell-cycle and proliferation regulation
ETS1	Transcriptional regulation, development, differentiation
FGFR1	Neurodevelopmental signaling, progenitor biology
OGT	Chromatin and transcription regulation through O-GlcNAc signaling
PHF8	Histone demethylase, active chromatin, neurodevelopment
RBPJ	NOTCH pathway transcriptional mediator
POLR2A	Active transcription, RNA polymerase II
KDM1A	Chromatin remodeling; also appears in McClay, so not fully NPC-only if McClay has signal



Study 1: Biological insights from 108 schizophrenia-associated genetic loci

Ripke

- **Among all TCF4 peaks, how many fall inside schizophrenia risk loci?**

The NPC TCF4 dataset has the highest proportion of peaks inside schizophrenia loci (5.78% in comparison with 4.22% and 4.06% for Dr. McClay and Forrest peaks, respectively)

- **Among the 108 schizophrenia loci, how many contain at least one TCF4 peak?**

Forrest and Dr. McClay dataset cover more schizophrenia loci overall (around 37% vs 24% for NPC), probably because they have many more total TCF4 peaks.

So, the difference is:

NPC: fewer total peaks, but higher percentage inside SCZ loci

McClay and Forrest: more total peaks, broader coverage of SCZ loci

12 out of 108 schizophrenia loci (~11%) contain TCF4 peaks in all three datasets

The most notable shared loci are:

1. chr10 locus: 104.4 to 105.2 Mb

1. Contains genes such as **AS3MT, CNNM2, CYP17A1, NT5C2**
2. NHGRI GWAS annotations include **ASD, ADHD, bipolar disorder, MDD, and schizophrenia**
3. This is one of the strongest schizophrenia-associated regions in the table.

2. chr12 locus: 123.4 to 123.9 Mb

1. Contains genes including **ABCB9, ARL6IP4, CDK2AP1, OGFOD2, SETD8**
2. NHGRI GWAS annotations include **schizophrenia**, HDL cholesterol, head circumference, and height.

3. chr11 locus: 46.3 to 46.7 Mb

1. Contains **AMBRA1, ARHGAP1, CHRM4, MDK, ZNF408**
2. NHGRI GWAS annotation includes **schizophrenia**
3. KO phenotype column also suggests neurological relevance for several genes.

Study 2: Mapping genomic loci implicates genes and synaptic biology in schizophrenia

Trubetskoy

- **LD-region overlap (313 LD region):** About 35 to 51% of schizophrenia GWAS LD regions contain at least one TCF4 binding site (34% for NPC, 39% for Dr. McClay, and 51% for Forrest)
- **Locus-level overlap with final SCZ loci (compared TCF4 peaks with the final autosomal schizophrenia loci from Trubetskoy Table S3- 282 SCZ regions):** A large fraction of schizophrenia loci overlap TCF4 binding sites. Forrest shows the highest coverage (~59%), followed by McClay (~40%), then NPC(~35%). However, this may partly reflect total peak number:
 - NPC has around 4,000 peaks
 - Dr. McClay and Forrest have around 11,000 peaks
- **Gene-level overlap within SCZ regions:** At the gene level, NPC has the highest percentage of its TCF4-associated genes located in SCZ regions (~8% in comparison with ~6%)
- **Shared genes across all three TCF4 datasets (NPC, Dr. McClay, and Forrest):**
 - TAOK2: neurodevelopmental kinase; located in the 16p11.2 neurodevelopmental risk locus
 - NTNG2: synaptic adhesion molecule involved in neuronal connectivity
 - ANKRD11: chromatin regulator linked to neurodevelopmental disorders and cognitive impairment
 - RERE: developmental transcriptional regulator involved in brain development
 - MSI2: RNA-binding protein involved in neural progenitor/stem cell regulation
 - SMPD3: sphingomyelin metabolism and synaptic signaling
 - ITPKB: calcium signaling regulator
 - NDRG4: neuronal survival and neurite growth
 - PSD3: synaptic signaling protein

Study 2: Mapping genomic loci implicates genes and synaptic biology in schizophrenia

Trubetskoy

- **S10b validation confirmed the S3 analysis:** we validate the findings using Table S10b, which contains schizophrenia loci defined by independent conditional association signals. 60% of SCZ risk loci overlaps with Forrest TCF4. dataset following 50.2% with Dr. McClay's and 38.68% with NPC's dataset.
- **51 of 243 S10b loci** were overlapped by TCF4 peaks in **all three datasets**, which equals 21%.
- **Overlap with Trubetskoy prioritized genes:** overlap of 120 prioritized genes with NPC TCF4 (14.2%) , Dr. McClay TCF4 (29.2%) and Forrest TCF4 (32.5%). Forrest and Dr. McClay capture more prioritized genes, again likely partly because they contain more total TCF4-associated genes.

Study 3: Single-cell multi-cohort dissection of the schizophrenia transcriptome

Ruzicka

- NPC TCF4 targets are enriched mainly in **glial**, vascular, and **inhibitory neuronal** populations, while the SH-SY5Y-based datasets, especially Forrest, are enriched more strongly in **excitatory cortical neuron** subclasses.
- Forrest TCF4 shows the strongest overlap with schizophrenia DE genes from Ruzicka, especially in excitatory neuron populations.
- The directional analysis shows that Forrest TCF4 has the strongest enrichment with schizophrenia DE genes, especially in SCZ-upregulated excitatory and inhibitory neuronal cell types. McClay has limited but significant enrichment, while NPC does not show FDR-significant enrichment in S4.
- The reproducible schizophrenia DE genes again support the strongest signal in **Forrest TCF4**, especially in excitatory cortical neuron subclasses. McClay also shows one significant excitatory neuron signal, while NPC does not show reproducible DE enrichment.

Study 4: Deconvolution of transcriptional networks identifies TCF4 as a master regulator in schizophrenia

Doostparast

We had 2 cell lines:

- 1.Day 3 neural progenitor cells, NPCs
- 2.Day 14 glutamatergic neurons

Dataset	OR	P-value
NPC	1.71	8.95×10^{-39}
Dr. McClay	1.41	9.02×10^{-20}
Forrest	1.69	1.02×10^{-39}

- For Day 3 NPCs, all three TCF4 datasets significantly overlapped TCF4 knockdown DE genes
- When Day 3 NPC DE genes were separated into upregulated and downregulated genes, each TCF4 dataset showed a different direction pattern.

Dataset	Stronger direction	Interpretation
NPC TCF4	Upregulated	NPC peaks overlap more genes that increase after TCF4 KD
Dr. McClay TCF4	Downregulated	Dr. McClay peaks overlap more genes that decrease after TCF4 KD
Forrest TCF4	Strongly downregulated	Forrest peaks show very strong enrichment among genes decreased after TCF4 KD

A similar pattern was observed in the comparison with **Blake’s dataset**, where NPC peaks showed greater overlap with genes upregulated after TCF4 knockdown. (OR = 1.24, p-val= 8.44×10^{-4})

- Shared directional overlap across datasets:** The strongest reproducibility is in the downregulated gene overlaps, especially between McClay and Forrest. The fact that opposite-direction comparisons showed zero shared genes which means same genes are not directionally different between datasets.
- The enrichment is still significant in Day 14 glutamatergic neurons, but it is generally **slightly weaker than in Day 3 NPCs**. TCF4 is highly active during early neurodevelopment, particularly in NPCs, but remains transcriptionally significant in glutamatergic neurons.

Dataset	Day 3 OR	Day 14 OR
NPC	1.71	1.45
Dr. McClay	1.41	1.46
Forrest	1.69	1.51

Study 4: Deconvolution of transcriptional networks identifies TCF4 as a master regulator in schizophrenia

Doostparast

- In Day 14 glutamatergic neurons, the directional patterns changed.

TCF4 dataset	Day 3 NPC	Day 14 Glut	Impression
NPC	Up > Down	Up $\approx$ Down	Very stable
Dr. McClay	Down only	Up only	Complete reversal
Forrest	Strong Down	Moderate Down	Same trend, weaker

Pathway analysis:

- **NPC peaks** capture early progenitor biology, including RNA splicing, translation/ribosome biology, DNA/chromosome organization, and cell-cycle processes.
- **Dr. McClay and Forrest peaks** capture more neuronal and neurite-development biology, especially Axanogenesis and neuron projection development.
- When TCF4 is knocked down in NPCs (Day 3), genes associated with NPC TCF4 peaks (our dataset) that go up are mainly related to **RNA processing and protein synthesis machinery**. This may suggest that TCF4 normally **represses or constrains some RNA-processing/translation-related programs** in NPCs.
- Genes associated with Dr. McClay and Forrest peaks that decrease after TCF4 knockdown are enriched for **neuronal development, axon guidance, and synaptic organization**. This supports the idea that TCF4 may positively regulate neuronal maturation or neuronal projection programs.
- **Day 14 glutamatergic neuron:** NPC peaks still retain some progenitor-like/cell-cycle signature even in Day 14 glutamatergic neurons, but neuronal projection terms also appear. Dr. McClay and Forrest remain more similar to each other and continue to highlight neuronal structure, axon development, cytoskeleton, and synaptic organization.
- NPC peak-associated genes that increase after TCF4 knockdown in Day 14 neurons are linked to **adhesion, cytoskeleton, protein turnover, and synaptic vesicle/endocytosis pathways**. However, NPC peak-associated genes that decrease after TCF4 knockdown in Day 14 neurons are enriched for **cell-cycle, DNA replication, chromosome segregation, and RNA splicing pathways**.
- **Cross-stage comparison: Day 3 NPC versus Day 14 glutamatergic neurons using NPC peaks:** NPC TCF4 peaks carry a mixed signature: early RNA-processing/cell-cycle biology + neuronal-development pathways. As cells mature toward glutamatergic neurons, the neuronal projection and axon-related programs become more prominent.

Study 4: Deconvolution of transcriptional networks identifies TCF4 as a master regulator in schizophrenia

Doostparast

- **Overall CD09 enrichment with TCF4 peak-associated genes:** NPC peaks do not show overall enrichment in CD09, while McClay and Forrest remain significantly enriched. However, it was weak.
- When I separated CD09 DEGs into upregulated and downregulated genes:

- The most consistent dataset-specific pattern is **Forrest**:
 - Day 3 NPC: strong downregulated enrichment
 - Day 14 SCZ glutamatergic neurons: moderate downregulated enrichment
 - CD09 healthy glutamatergic neurons: strong downregulated enrichment
- NPC peaks are different:
 - Strong in Day 3 NPC (especially upregulation)
 - Still significant in Day 14 SCZ glutamatergic neurons (Up $\approx$ Down)
 - Not enriched overall in CD09 healthy glutamatergic neurons

TCF4 dataset	Direction	OR	P-value	Interpretation
NPC peaks	Up	0.80	4.36×10^{-3}	Depleted, not enriched
NPC peaks	Down	1.29	2.84×10^{-3}	Modest down enrichment
Dr. McClay peaks	Up	1.25	6.39×10^{-4}	Modest up enrichment
Dr. McClay peaks	Down	1.27	3.15×10^{-3}	Modest down enrichment
Forrest peaks	Up	1.05	0.520	Not enriched
Forrest peaks	Down	1.77	1.56×10^{-12}	Strongest CD09 signal

RNA-seq dataset	NPC peaks	Dr. McClay peaks	Forrest peaks
Day 3 NPC	OR 1.71	OR 1.41	OR 1.69
Day 14 SCZ glut	OR 1.45	OR 1.46	OR 1.51
CD09 healthy glut	OR 0.98	OR 1.29	OR 1.34

getting weaker

Study 6: nFGFR1/NOTCH ChIP-seq in hiPSC-derived NPCs from Control and Schizophrenia patients

GSE92873

- At the gene level, the NPC overlap is also very high
- so based on peak-level and gene-level analysis:** NPC TCF4 is strongly connected to FGFR1 and NOTCH regulatory programs.
- Four-way relationship: NPC TCF4, CTCF, FGFR1, and NOTCH:**
 - The four-way peak overlap is:
NPC TCF4 + CTCF + Patient FGFR1 + Patient NOTCH = 854 peaks
This represents:
21.65% of NPC TCF4 peaks
79.89% of NPC-CTCF overlapping peaks
 - The four-way gene overlap is:
NPC TCF4 + CTCF + Patient FGFR1 + Patient NOTCH = 2,241 genes
This represents:
60.47% of NPC TCF4 genes
81.67% of NPC-CTCF genes

This is a large shared regulatory space.

This suggests that NPC TCF4, CTCF, FGFR1, and NOTCH may converge on a shared neurodevelopmental gene regulatory program in SCZ patients.

gene-level comparison:

Comparison	Overlap Genes	First Gene Set Size	Second Gene Set Size	% of First Gene Set	% of Second Gene Set
Dr. McClay vs Control_FGFR1_genes	3,349	6,174	14,128	54.24%	23.70%
Dr. McClay vs Patient_FGFR1_genes	4,271	6,174	18,329	69.18%	23.30%
Dr. McClay vs Patient_NOTCH_genes	3,126	6,174	14,207	50.63%	22.00%
NPC vs Control_FGFR1_genes	3,338	3,706	14,128	90.07%	23.63%
NPCs vs Patient_FGFR1_genes	3,441	3,706	18,329	92.85%	18.77%
NPC vs Patient_NOTCH_genes	3,091	3,706	14,207	83.41%	21.76%
Forrest_genes vs Control_FGFR1_genes	2,998	6,048	14,128	49.57%	21.22%
Forrest_genes vs Patient_FGFR1_genes	3,895	6,048	18,329	64.40%	21.25%
Forrest_genes vs Patient_NOTCH_genes	3,200	6,048	14,207	52.91%	22.52%

I used universal background gene for enrichment analysis

Comparison	Overlap	Odds Ratio	P-value
Dr. McClay vs Control FGFR1	2666	2.17	3.1×10^{-115}
Dr. McClay vs Patient FGFR1	3249	2.69	2.4×10^{-167}
Dr. McClay vs Patient NOTCH	2419	1.79	5.1×10^{-67}
NPC vs Control FGFR1	2566	20.43	$< 2.2 \times 10^{-308}$
NPC vs Patient FGFR1	2609	16.46	$< 2.2 \times 10^{-308}$
NPC vs Patient NOTCH	2365	9.76	$< 2.2 \times 10^{-308}$
Forrest vs Control FGFR1	2290	1.99	3.5×10^{-83}
Forrest vs Patient FGFR1	2805	2.44	7.5×10^{-125}
Forrest vs Patient NOTCH	2299	2.21	1.2×10^{-109}

Study 7: H3K27ac (active enhancer/promoter mark) in H9-derived NPCs

GSE96178

- TCF4 binding in neural progenitor cells is strongly concentrated within active enhancer and promoter elements.
- TCF4 regulatory architecture is highly cell-type dependent

Dataset	Total TCF4 Peaks	H3K27ac Peaks	Overlapping TCF4 Peaks	% of TCF4 Peaks Overlapping H3K27ac	Overlapping H3K27ac Peaks	% of H3K27ac Peaks Overlapping TCF4
NPC	3,945	48,258	2,704	68.54%	3,333	6.91%
Dr. McClay	11,246	48,258	416	3.70%	412	0.85%
Forrest	9,999	48,258	815	8.15%	866	1.80%

Dataset	Total TCF4 Peaks	H3K27ac Peaks	Overlapping TCF4 Peaks	% of TCF4 Peaks Overlapping H3K27ac	Overlapping H3K27ac Peaks	% of H3K27ac Peaks Overlapping TCF4
NPC	3,945	48,258	2,704	68.54%	3,333	6.91%
Dr. McClay	11,246	48,258	416	3.70%	412	0.85%
Forrest	9,999	48,258	815	8.15%	866	1.80%

Study 8: H3K4me3 ChIP-seq peaks from H9-derived human NPCs
GSE176914

Dataset	% TCF4 genes associated with H3K4me3
NPC TCF4	87.51%
Dr. McClay TCF4	18.03%
Forrest TCF4	14.35%

Dataset	% TCF4 peaks overlapping H3K4me3
NPC TCF4	85.93%
Dr. McClay TCF4	11.04%
Forrest TCF4	9.50%

- In neural progenitor cells, TCF4 binding is highly concentrated at active promoters rather than being broadly distributed across distal regulatory regions.
- Dr. McClay and Forrest TCF4 peaks appear much less connected to the NPC promoter landscape, suggesting they reflect a **different cell-type-specific regulatory architecture**.
- Together with the H3K27ac result, this suggests: NPC TCF4 peaks are embedded in active chromatin, especially **active promoter regions**.

Study 9: ChIP-Seq analysis of H3K27me3 (repressive histone mark) in hN cells (H9-derived neurons / hN cells)
GSM772787

NPC TCF4 binding is strongly associated with **active chromatin marks**, especially H3K4me3 and H3K27ac, and shows **minimal overlap with the repressive mark H3K27me3**.

Dataset	% TCF4 peaks overlapping H3K27me3	H3K27me3-associated TCF4 genes
NPC TCF4	1.19%	47
Dr. McClay TCF4	10.68%	884
Forrest TCF4	4.66%	391

Study 10: EP300 / p300 ChIP-seq in human neural cells
GSE127584

- NPC TCF4 peaks are not simply overlapping random neural/NPC chromatin regions. They preferentially overlap active regulatory elements.

NPC TCF4 is preferentially localized to **active promoters** and **enhancers** and is largely excluded from Polycomb-repressed chromatin.

Mark	Cell Type	NPC TCF4	Dr. McClay TCF4	Forrest TCF4
H3K4me3	NPC	85.9%	11.0%	9.5%
H3K27ac	NPC	68.5%	3.7%	8.2%
EP300	Neurons	62.6%	1.8%	6.9%
H3K27me3	NPC	1.2%	10.7%	4.7%

Mark	NPC Genes	Dr. McClay Genes	Forrest Genes
H3K4me3	3243	1113	868
H3K27ac	2563	374	727
EP300	2404	195	646
H3K27me3	47	884	391

Dataset	% TCF4 peaks overlapping EP300
NPC TCF4 vs EP300	62.2%
Dr. McClay TCF4 vs EP300	1.6 to 1.8%
Forrest TCF4 vs EP300	6.8 to 6.9%

EP300 Peak Set	Dataset	Gene overlaps
Optimal	NPC TCF4	2,404
Optimal	Dr. McClay TCF4	195
Optimal	Forrest TCF4	646
Conservative	NPC TCF4	2,387
Conservative	Dr. McClay TCF4	176
Conservative	Forrest TCF4	642

Mark	Meaning	NPC TCF4 overlap
H3K4me3	Active promoter/TSS	85.9%
H3K27ac	Active enhancer/promoter	68.5%
EP300	Active enhancer/co-activator	62.6%
H3K27me3	Repressive/Polycomb chromatin	1.2%

Study 11: regulatory genomic profiling from iPSC-derived human cortical neurons and postmortem brain.

GSE79965

iPSC-derived human cortical neurons + postmortem brain

- **Peak-level overlap with H3K27ac, ATAC, and CHD8:** NPC TCF4 peaks represent active neurodevelopmental regulatory elements more clearly than the SH-SY5Y datasets
- Dr. McClay and Forrest show much lower overlap with these features

Feature	% NPC TCF4 peaks overlapping
Neuron ATAC-seq	77.50%
Neuron H3K27ac	74.94%
Brain CHD8	65.67%
Neuron CHD8	46.72%

- **TAD-level analysis**
- Of the 1,544 TADs containing NPC TCF4 peaks, 1,483 also contain H3K27ac, ATAC, and CHD8 signal. That equals about **96.05%** of NPC TCF4-containing TADs: **Almost every NPC TCF4-containing TAD also contains active chromatin and CHD8 signal.**

** CHD8 is a major autism/neurodevelopmental regulator and ATAC/H3K27ac mark active chromatin*

Category	TAD count	% of all TADs
TADs with NPC TCF4	1,544	49.30%
TADs with Dr. McClay TCF4	1,854	59.20%
TADs with Forrest TCF4	2,135	68.17%
NPC TCF4 + H3K27ac + ATAC + CHD8 TADs	1,483	47.35%

- at the **peak level**, datasets look very different.
At the **TAD level**, they share many regulatory neighborhoods
- Exact peak positions may differ across different cell types, but they may still fall within the **same regulatory domains**

Comparison	Shared TADs	% of First TAD Set	% of Second TAD Set
NPC TADs vs Dr. McClay TADs	1,110	71.89%	59.87%
NPC TADs vs Forrest TADs	1,206	78.11%	56.49%
Dr. McClay TADs vs Forrest TADs	1,484	80.04%	69.51%
NPC vs Dr. McClay vs Forrest TADs	941	60.95% of NPC TADs	—

Study 11: regulatory genomic profiling from iPSC-derived human cortical neurons and postmortem brain.

GSE79965

- **TAD-level analysis:** NPC TCF4 peak density across TADs tracks most strongly with CHD8 peak density, especially brain CHD8. In a simpler way, the top-100 TAD analysis shows that NPC TCF4 is especially associated with **CHD8-rich TADs**.
- I identified **TAD_1375**, located at approximately **chr19:57.81 to 58.98 Mb**, as the only TAD simultaneously in the top 100 for:
 - NPC TCF4
 - H3K27ac
 - ATAC
 - neuron CHD8
 - brain CHD8

it contains one of the largest **KRAB zinc-finger gene clusters** in the human genome, with about **46 zinc finger-related genes** among 64 genes/transcripts.

- **RNA-seq expression analysis:** Genes near NPC TCF4 binding sites tend to be more actively transcribed in iPSC-derived cortical neurons.

For NPC TCF4:

Feature	Shared top-100 TADs
Brain CHD8	50%
Neuron CHD8	49%
H3K27ac	16%
ATAC	6%

The Spearman correlation also supports this:

Feature	NPC TCF4 correlation
Brain CHD8	0.594
Neuron CHD8	0.522
H3K27ac	0.391
ATAC	0.230

TCF4-associated genes expressed in GSE79965 RNA-seq

Dataset	Total TCF4 genes	Expressed genes FPKM > 0	Expressed genes FPKM > 1	% expressed FPKM > 1
NPC TCF4	3,650	2,795	2,352	64.44%
Dr. McClay TCF4	5,671	3,756	2,207	38.92%
Forrest TCF4	5,450	3,425	2,171	39.84%

Study 12: Knockdown of Human TCF4 Affects Multiple Signaling Pathways Involved in Cell Survival, Epithelial to Mesenchymal Transition and Neuronal Differentiation (2013)

Forrest, Blake

TCF4 dataset	New Blake DEG set	Background genes	TCF4 genes in background	Blake genes in background	Overlap genes	Odds ratio	P-value
NPC_TCF4	New Blake all FDR < 0.05	18,342	2,365	5,957	761	0.98	0.639
NPC_TCF4	New Blake up FDR < 0.05	18,342	2,365	1,873	286	1.25	8.44e-04
NPC_TCF4	New Blake down FDR < 0.05	18,342	2,365	4,108	476	0.86	0.998
NPC_TCF4	New Blake all FDR < 0.01	18,342	2,365	2,516	331	1.03	0.346
NPC_TCF4	New Blake up FDR < 0.01	18,342	2,365	869	132	1.22	0.0236
NPC_TCF4	New Blake down FDR < 0.01	18,342	2,365	1,647	199	0.92	0.858
Dr. McClay_TCF4	New Blake all FDR < 0.05	18,342	3,348	5,957	1,094	1.01	0.400
Dr. McClay_TCF4	New Blake up FDR < 0.05	18,342	3,348	1,873	316	0.90	0.953
Dr. McClay_TCF4	New Blake down FDR < 0.05	18,342	3,348	4,108	786	1.08	0.0515
Dr. McClay_TCF4	New Blake all FDR < 0.01	18,342	3,348	2,516	444	0.95	0.809
Dr. McClay_TCF4	New Blake up FDR < 0.01	18,342	3,348	869	143	0.88	0.928
Dr. McClay_TCF4	New Blake down FDR < 0.01	18,342	3,348	1,647	301	1.00	0.501
Forrest_TCF4	New Blake all FDR < 0.05	18,342	2,855	5,957	943	1.03	0.253
Forrest_TCF4	New Blake up FDR < 0.05	18,342	2,855	1,873	366	1.36	6.35e-07
Forrest_TCF4	New Blake down FDR < 0.05	18,342	2,855	4,108	583	0.87	0.997
Forrest_TCF4	New Blake all FDR < 0.01	18,342	2,855	2,516	441	1.18	0.0021
Forrest_TCF4	New Blake up FDR < 0.01	18,342	2,855	869	181	1.46	1.44e-05
Forrest_TCF4	New Blake down FDR < 0.01	18,342	2,855	1,647	260	1.02	0.409

Old Blake = Using the .txt gene files Dr. McClay provided (published Blake’s DGE list)

New Blake = Gene lists obtained from reanalysis of Blake’s raw microarray data

- Forrest_TCF4 shows the strongest and most consistent enrichment, while NPC_TCF4 shows a weaker upregulated-gene signal.

TCF4 gene set	Blake DEG threshold	Blake DEG set	Overlap genes	Odds ratio	P-value
TCF4 all peaks	FDR < 0.05	Upregulated	286	1.247	8.44e-04
TCF4 motif peaks	FDR < 0.05	Upregulated	229	1.316	2.14e-04
TCF4 promoter + E-box	FDR < 0.05	Upregulated	200	1.342	2.02e-04
TCF4 all peaks	FDR < 0.01	Upregulated	132	1.222	0.0236
TCF4 motif peaks	FDR < 0.01	Upregulated	101	1.217	0.0426

Study 12: Knockdown of Human TCF4 Affects Multiple Signaling Pathways Involved in Cell Survival, Epithelial to Mesenchymal Transition and Neuronal Differentiation (2013)

Forrest, Blake

TCF4 dataset	Blake DEG set	Background genes	TCF4 genes in background	Blake genes in background	Overlap genes	Odds ratio	P-value
NPC_TCF4	Old Blake all DEGs	18,342	2,365	1,126	166	1.18	0.0328
NPC_TCF4	Old Blake upregulated	18,342	2,365	469	85	1.51	6.61e-04
NPC_TCF4	Old Blake downregulated	18,342	2,365	658	81	0.95	0.693
Dr. McClay_TCF4	Old Blake all DEGs	18,342	3,348	1,126	212	1.04	0.315
Dr. McClay_TCF4	Old Blake upregulated	18,342	3,348	469	84	0.98	0.596
Dr. McClay_TCF4	Old Blake downregulated	18,342	3,348	658	128	1.08	0.222
Forrest_TCF4	Old Blake all DEGs	18,342	2,855	1,126	224	1.38	3.54e-05
Forrest_TCF4	Old Blake upregulated	18,342	2,855	469	95	1.39	0.00356
Forrest_TCF4	Old Blake downregulated	18,342	2,855	658	129	1.34	0.00270

- Using the published Blake DEG lists, Forrest_TCF4 genes are enriched across all Blake DEG categories, while NPC_TCF4 genes are specifically enriched among genes upregulated after TCF4 knockdown.

Old Blake = Using the .txt gene files that Dr. McClay provided (published Blake's DGE list)

New Blake = Gene lists obtained from reanalysis of Blake's raw microarray data

Study 13: Transcriptome-wide isoform-level dysregulation in ASD, schizophrenia, and bipolar disorder

Gandal

- **SCZ**
- All three TCF4 gene sets overlap significantly with psychiatric disorder DGE genes from Gandal, but Dr. McClay and Forrest are consistently stronger than NPC.
- Both upregulated and downregulated SCZ genes are enriched, but the **upregulated** direction is stronger
- **ASD and BD**
- All three datasets are significant, and Dr. McClay/Forrest are again stronger
- The ASD directional table shows both up and down enrichment, but the **upregulated genes are stronger**, especially for Forrest and Dr. McClay.

Disorder	Dataset	Overlap genes	Odds ratio	P-value
ASD	NPC	256	1.56	1.91e-09
ASD	Dr. McClay	367	1.82	4.71e-20
ASD	Forrest	342	1.85	3.99e-20
BD	NPC	154	1.29	4.73e-03
BD	Dr. McClay	264	1.88	1.28e-16
BD	Forrest	193	1.39	8.42e-05

Dataset	SCZ overlap	OR	P-value
NPC	660	1.35	6.05e-10
Dr. McClay	1,012	1.78	2.09e-42
Forrest	910	1.72	4.88e-35

Direction	Dataset	OR
SCZ Up	NPC	1.36
SCZ Up	Dr. McClay	1.78
SCZ Up	Forrest	1.71
SCZ Down	NPC	1.25
SCZ Down	Dr. McClay	1.54
SCZ Down	Forrest	1.50

Disorder	Direction	Dataset	Overlap genes	Odds ratio	P-value
ASD	Up	NPC	131	1.74	8.66e-08
ASD	Up	Dr. McClay	173	1.82	1.27e-10
ASD	Up	Forrest	161	1.85	1.98e-10
ASD	Down	NPC	125	1.37	2.01e-03
ASD	Down	Dr. McClay	194	1.74	2.43e-10
ASD	Down	Forrest	181	1.78	1.81e-10
BD	Up	NPC	110	1.46	5.79e-04
BD	Up	Dr. McClay	188	2.14	1.31e-16
BD	Up	Forrest	116	1.27	2.21e-02
BD	Down	NPC	44	0.99	1.00
BD	Down	Dr. McClay	76	1.40	1.19e-02
BD	Down	Forrest	77	1.58	5.82e-04